

The additive group and a connective descendability example

Let $k = \mathbf{F}_2$ and $G = \mathbf{G}_a$. The dilation action grades $\mathcal{O}(G) = k[t]$. Its restricted dual

$$A := \mathcal{O}(G)^\circ = \bigoplus_{d \geq 0} (kt^d)^\vee$$

is the graded distribution algebra of G . The heart $\mathrm{QCoh}(BG)^\heartsuit$ identifies with the A -modules on which every element is killed by all but finitely many distribution operators. If $e_d = (t^d)^\vee$, then $e_a e_b = \binom{a+b}{a} e_{a+b}$. Hence, by binary expansion,

$$A \simeq k[x_0, x_1, \dots] / (x_i^2), \quad x_i = e_{2^i}.$$

The regular representation $M := \mathcal{O}(G)$ is an A -module by

$$x_i(t^d) = \binom{d}{2^i} t^{d-2^i}.$$

Equivalently, writing $m_F = t^{\sum_{i \in F} 2^i}$ for finite $F \subset \mathbf{N}$, one has

$$x_i m_F = \begin{cases} m_{F \setminus \{i\}}, & i \in F, \\ 0, & i \notin F. \end{cases}$$

Proposition. As objects of Mod_A ,

$$0 \neq M, \quad k \otimes_A^{\mathbf{L}} M \simeq 0, \quad M \in \mathrm{Loc}_A(k);$$

consequently $M \otimes_A^{\mathbf{L}} M \simeq 0$.

Proof. The finite-dimensional subrepresentations $\langle 1, t, \dots, t^N \rangle$ exhaust M . Each factors through a finite-dimensional local quotient of A , so it has a finite filtration with successive quotient k . Thus $M \in \mathrm{Loc}_A(k)$.

Put

$$A_r = k[x_0, \dots, x_r] / (x_i^2), \quad M_r = \langle m_F : F \subseteq \{0, \dots, r\} \rangle,$$

and define $A_{>r}, M_{>r}$ similarly. Then $A = A_r \otimes_k A_{>r}$, $M = M_r \otimes_k M_{>r}$, and $M_r \simeq A_r$ as A_r -modules. Since $k \simeq \mathrm{colim}_r A / (x_0, \dots, x_r)$, it follows that

$$k \otimes_A^{\mathbf{L}} M \simeq \mathrm{colim}_r M_{>r}.$$

The transition $M_{>r} \rightarrow M_{>r+1}$ sends m_F to $m_{F \setminus \{r+1\}}$ if $r+1 \in F$, and to 0 otherwise. Every element therefore dies after finitely many transitions, so this colimit is zero. Finally, $M \otimes_A^{\mathbf{L}} -$ kills k , hence kills $\mathrm{Loc}_A(k)$, which contains M . $\square$

Corollary. Let $S = HA$ and let $R = S \oplus HM \rightarrow S$ be the split square-zero extension. Both rings are discrete and the map is not an equivalence. For every $n \geq -1$, put $T_n := \mathrm{colim}_{S^n} S$, with the colimit formed in CAlg_R . The bar construction and $M \otimes_A^{\mathbf{L}} M = 0$ give

$$T_n \simeq S \oplus \Sigma^{n+1} HM.$$

The augmentation $T_n \rightarrow S$ is descendable. Indeed its fiber N is an S -module, hence is a retract of $S \otimes_{T_n} N$; thus N belongs to the thick tensor ideal generated by S , and the fiber sequence $N \rightarrow T_n \rightarrow S$ shows that this ideal contains the unit T_n .